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MTE 421: Mechatronics Systems Design

Course Project: Cooling and Sorting Automated
Chocolate Bars Production Line

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1 Introduction and Proposed Solution

In modern chocolate manufacturing, maintaining strict thermal profiles is critical to preserving product quality, preventing surface blooming, and ensuring structural integrity.

This report details the design, simulation, and analysis of an automated mechatronic production line designed to convey, cool (to a target of 20°C), weigh, and sort chocolate bars.

Proposed Innovative Solution:

The proposed system integrates a closed-loop variable-speed conveyor belt with dynamic forced-air cooling. Unlike traditional fixed-speed lines, my system utilizes a primary PID controller to adapt the conveyor's linear speed (nominally 0.1 m/s) and a secondary PID loop to modulate cooling fan speeds based on real-time surface temperature readings.

A multi-sensor inspection station —combining non-contact infrared (IR) sensors for temperature and dynamic load cells for weight (target 100 g)— acts as the primary feedback node. Defective products (out-of-spec weight, overheated, or under-cooled) are diverted using a high-speed pneumatic sorting mechanism, ensuring zero contamination of the accepted batch.

2 Component Selection and Justification

To fulfill the requirements of handling at least 100 chocolate bars continuously, components were selected based on dynamic load requirements, precision, and industrial reliability.

2.1 Mechanisms

The mechanical design of the system relies on a precise feeding, conveying, and robotic sorting architecture. The primary mechanisms are detailed below:

- **Rotary Feeder Mechanism:** A stepper motor uses a central shaft and bearings to rotate a high product disk. The disk features specific pockets where the chocolate bars sit, ensuring each product is perfectly aligned and stopped precisely over the drop-off point to enter the line one by one.

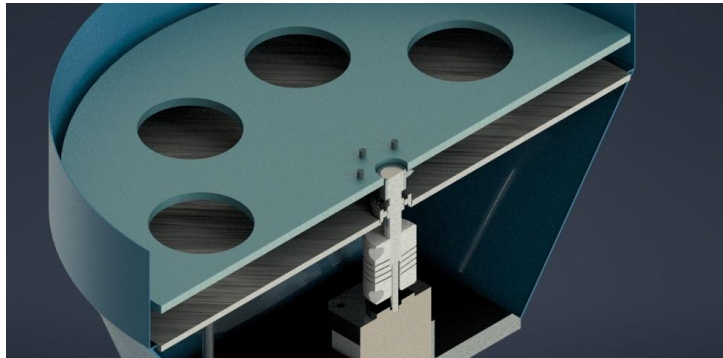


Figure 1: Feeding Motor and Shaft

- **Conveyor Belt:** The conveyor moves the products from the feeding mechanism through the cooling zone to the Cartesian robotic sorter. A stepper motor, connected to the drive roller using a flexible coupler, precisely controls the speed of the belt to allow for adequate cooling time.



Figure 2: Conveyor Belt

- **Inspection Setup:**

- **Temperature Sensor:** Mounted directly above the conveyor belt. It reads the surface temperature of the chocolate bar as it passes beneath the opening to ensure it has reached the target 20°C.
- **Shape/Weight Sensor:** Mounted at a fixed height above the belt, this sensor sends out a light beam to measure returning light, detecting variations in product height and dimensions to ensure structural integrity.

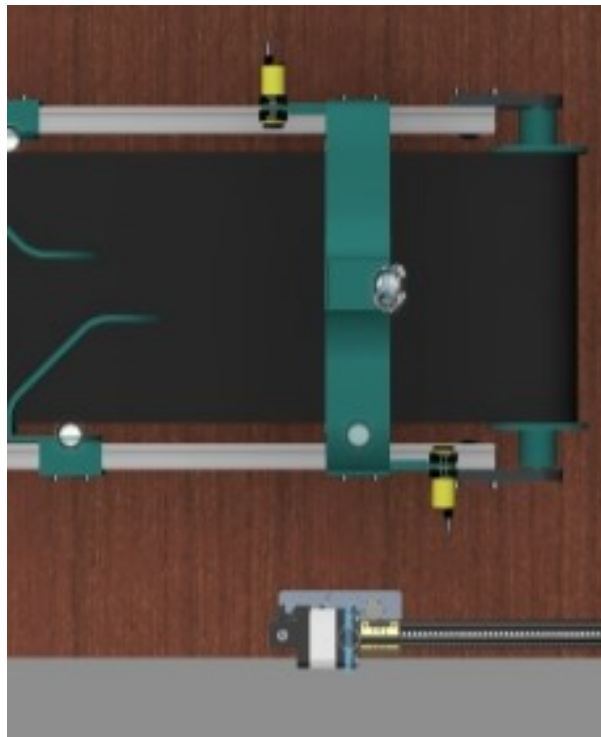


Figure 3: Inspection Setup

- **Cartesian Robotic Arm:**

A 3-axis gantry system performs the final sorting operations.

- **X-Axis:** Constructed from rigid aluminum extrusions with smooth rods acting as guides for stable linear motion. A motor-driven lead screw pushes the carriage, while timing belts and pulleys are utilized for rapid power transmission. Limit switches are installed at the physical ends of the track to prevent over-travel.

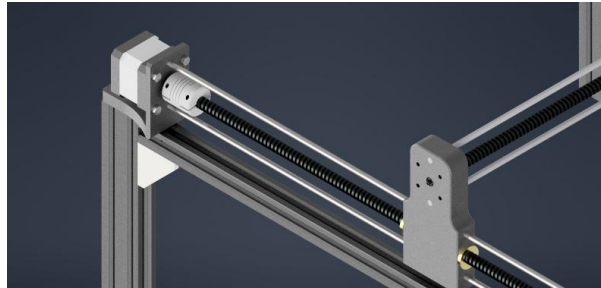


Figure 4: X-Axis: Lead Screw and Motor

- **Y-Axis:** Powered by a DC motor equipped with a linear encoder. The encoder provides feedback to ensure the carriage moves forward and backward and stops exactly at the required coordinates. The carriage converts the spinning of the lead screw into straight motion while supporting the entire Z-axis assembly. Limit switches secure the axis boundaries.

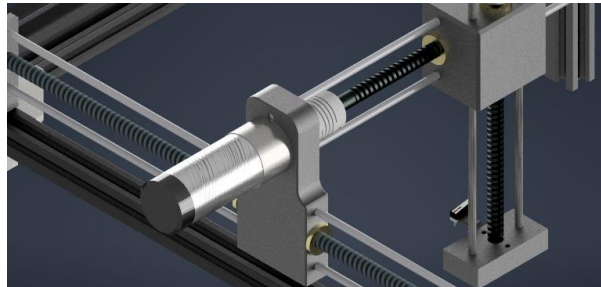


Figure 5: Y-Axis: DC Motor with Encoder

- **Z-Axis:** A stepper motor spins a vertical rod, smoothly connected via a flexible coupler to prevent shaking. This motion drives the end effector up and down.

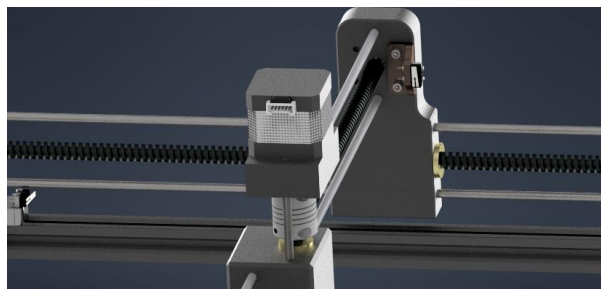


Figure 6: Z-Axis: Stepper Motor

- **The Gripper (End Effector):** A custom claw designed to handle the chocolate. A servo motor turns a drive gear, which synchronizes driven gears and linkages to convert the spin into a straight opening/closing motion for the fingers. The servo constantly checks its position, allowing the robot to accurately control the grip force and width so the chocolate is not crushed.



Figure 7: Servo Motor in Gripper

- **Base and Infrastructure:** The large X-Y-Z frame is anchored to a solid wooden base using standard base supports for stability.
 - **Storage Area:** Contains specific bins where the gripper places the chocolate bars based on their temperature and shape conditions. After the cooling stage, a sorting mechanism will classify the products into:
 - * **Accepted products:**
 - 1- Chocolate bars within the desired weight and cooling condition.
 - * **Rejected products:**
 - 2- Overheated
 - 3- Insufficiently Cooled Bars
 - 4- Improper Weight

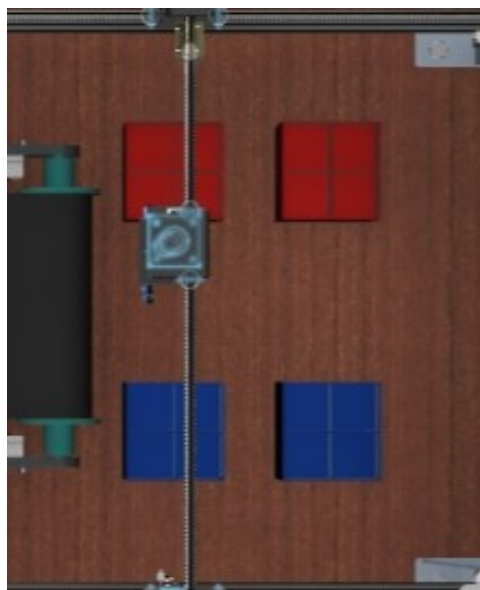


Figure 8: Storage Area

- **Control Box:** Houses the microcontrollers, motor drivers, power supply, emergency stop switch, and necessary circuitry to run the system.



Figure 9: Control Box

- **Final Production Line:**

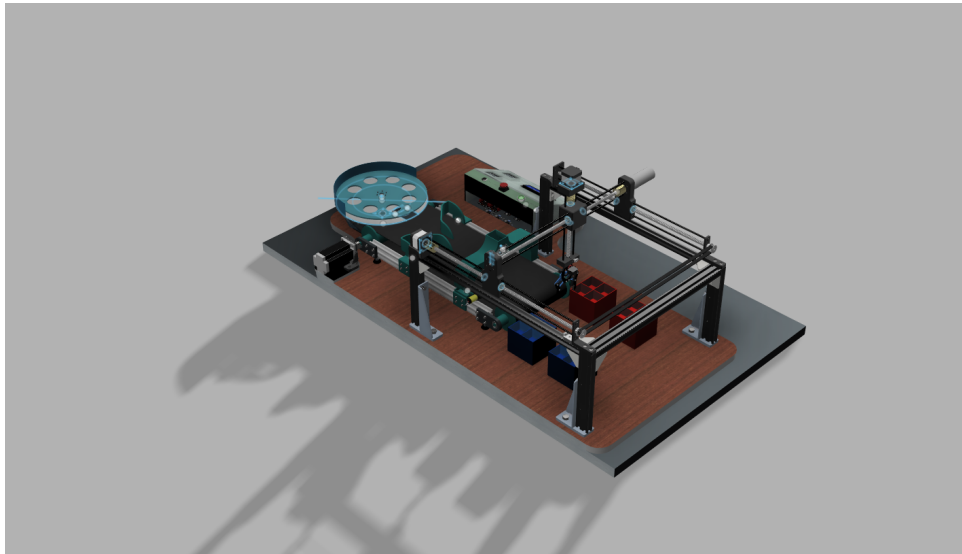


Figure 10: Final Production Line Isometric View

2.2 Electronics (Sensors)

- **Temperature Sensor:** An MLX90614 Non-Contact Infrared (IR) Thermometer. Contact sensors would damage the soft chocolate. The IR sensor provides an accuracy of $\pm 0.5^{\circ}\text{C}$ and rapid response times suitable for moving targets.
- **Weight Sensor:** A dynamic load cell paired with an HX711 24-bit ADC amplifier. The load cell is integrated into a floating section of the conveyor bed to measure the mass of the moving bar ($m \approx 0.1 \text{ kg}$).

2.3 Computing, Control, and Wiring Architecture

The electronic circuit forms the central nervous system of the automated chocolate production line, safely bridging the power supply, microcontrollers, sensor feedback arrays, and heavy-duty actuators.

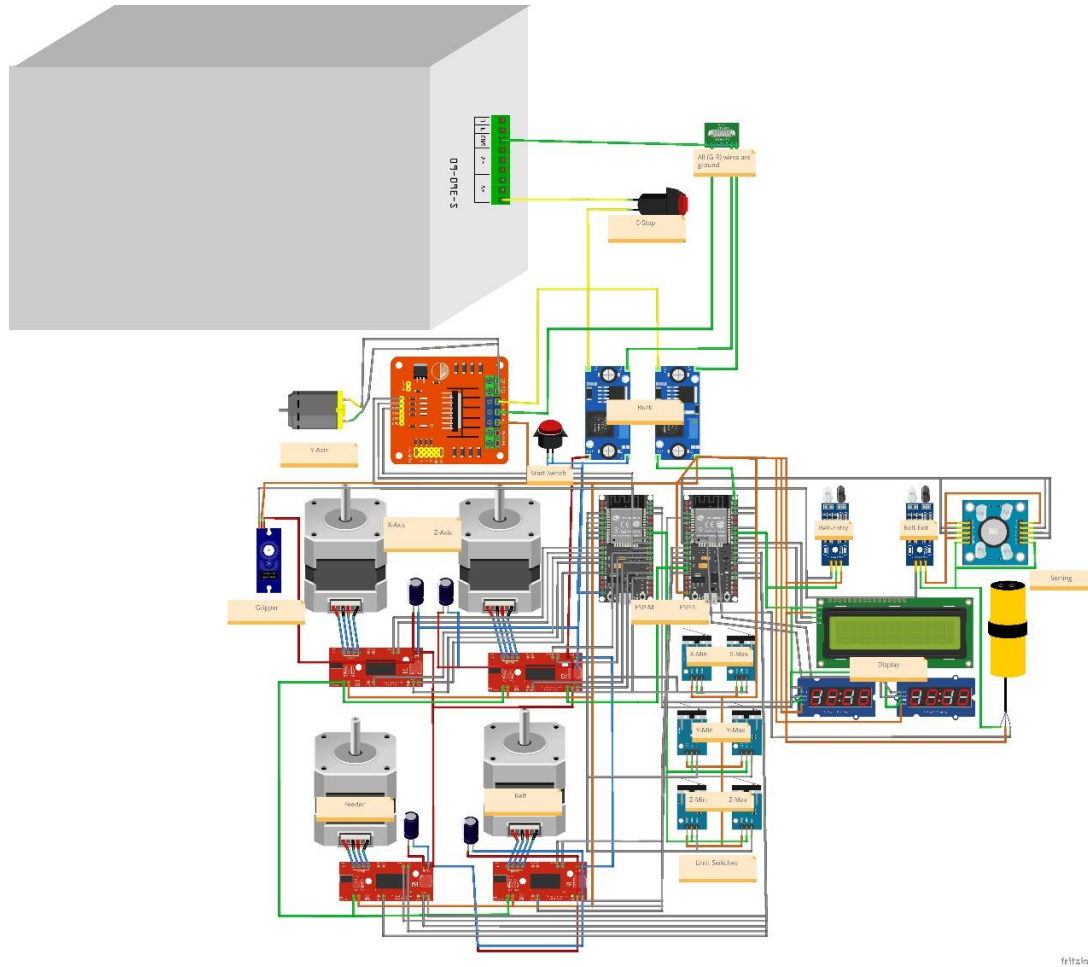


Figure 11: Wiring schematic demonstrating the integration of the ESP-based controllers, power distribution, and peripheral mechatronic components.

Control Logic and Processing: An ESP-based microcontroller architecture is utilized as the core computing unit. By utilizing a dual-controller setup, the computing load is distributed efficiently: one controller handles the continuous, high-frequency PID calculations for conveyor speed and dynamic fan cooling, while the other processes sensor data (temperature and weight) and triggers the high-speed pneumatic sorting actuator.

Sensor Integration and Actuator Control:

- **Actuator Drivers:** The main 24V DC gear motor (conveyor drive) and the 12V BLDC cooling fans are controlled using dedicated high-power motor drivers (H-bridge modules) receiving PWM signals from the ESP controllers. MOSFET module is used to actuate the 24V pneumatic solenoid valve for the diversion mechanism.
- **Feedback Loop:** The system reads real-time analog and digital signals from the MLX90614 (IR Temperature) and the HX711 ADC (Load cell/Weight). Additional photoelectric or IR break-beam sensors are positioned at the belt's entry and exit to track the physical location of the chocolate bars, ensuring precise timing for the sorting diverter.

Power Distribution and Safety:

- **Voltage Regulation:** Power is delivered by a main industrial power supply (24V DC). To accommodate components with varying voltage requirements, step-down buck converters are heavily utilized to provide stable, regulated 12V for the cooling fans and 5V/3.3V rails for the ESP controllers, logic sensors, and displays. All components share a common ground to ensure stable signal referencing and reliable operation.
- **Safety Mechanisms:** A physical Start switch initiates the automated sequence, while a hardware Emergency Stop (E-Stop) button is wired to cut power to the main actuators instantly, ensuring safe system operation during a mechanical jam or fault.
- **User Interface:** A central LCD screen and supplementary digital indicators provide operators with basic, real-time system status information, including current belt speed, average chocolate temperature, and fault codes.

2.4 Motor and Motor Drive

- **Main Conveyor Motor:** A 24V DC brushed gear motor. DC motors, when paired with high-reduction gearboxes, provide excellent low-speed torque characteristics.
- **Motor Drive:** An industrial H-Bridge MOSFET driver (e.g., Cytron MD30C) capable of handling continuous currents up to 30A. It accepts high-frequency PWM signals from the computing unit to seamlessly adjust the conveyor speed.
- **Cooling Fans:** 12V Brushless DC (BLDC) fans driven by standard PWM controllers, allowing the system to modulate airflow (convective heat transfer) based on the temperature error.

3 Force Analysis and Detailed Design

To correctly size the main conveyor motor, a dynamic load analysis is required. The system must transport at least 100 chocolate bars simultaneously.

System Parameters:

- Mass of a single chocolate bar: $m_{bar} = 0.1$ kg
- Total load mass: $m_{load} = 100 \times 0.1 = 10$ kg
- Estimated mass of the belt: $m_{belt} = 5$ kg
- Total moving mass: $m_{total} = 15$ kg
- Coefficient of kinetic friction (PU belt on steel bed): $\mu_k = 0.3$
- Drive roller radius: $r = 0.05$ m
- Nominal linear velocity: $v = 0.1$ m/s

Required Force and Torque:

The frictional force F_f opposing the motion is calculated as:

$$F_f = \mu_k \cdot N = \mu_k \cdot m_{total} \cdot g \quad (1)$$

$$F_f = 0.3 \cdot 15 \text{ kg} \cdot 9.81 \text{ m/s}^2 = 44.145 \text{ N} \quad (2)$$

The required torque τ at the drive roller is:

$$\tau = F_f \cdot r = 44.145 \text{ N} \cdot 0.05 \text{ m} = 2.207 \text{ Nm} \quad (3)$$

To ensure reliability and account for inertial spikes during startup, a safety factor (SF) of 2 is applied, yielding a design torque of 4.41 Nm.

Motor Speed Requirement:

The angular velocity ω of the roller to achieve 0.1 m/s is:

$$\omega = \frac{v}{r} = \frac{0.1}{0.05} = 2 \text{ rad/s} \quad (4)$$

Converting to Revolutions Per Minute (RPM):

$$\text{RPM} = \frac{\omega \cdot 60}{2\pi} = \frac{2 \cdot 60}{2\pi} \approx 19.1 \text{ RPM} \quad (5)$$

Motor Selection: Based on these calculations, a 24V DC Gear Motor with a rated torque of 5.0 Nm and a no-load speed of 30 RPM was selected.

Attached Document: Please refer to the attached file

Force_Analysis_and_Detailed_Design.pdf

which contains the detailed Actuator Sizing, Proper Material Selection and Manufacturing Processes, and Stress analysis.

4 System Schematic and Simscape Representation

The system architecture relies on two primary closed loops. The primary loop regulates the conveyor speed via a PID controller acting on the DC motor driver. The secondary loop monitors the IR temperature sensor and modulates the BLDC cooling fans.

In MATLAB Simscape, the physical network approach was utilized. The model bridges the electrical domain (DC voltage source, H-bridge), mechanical rotational domain (motor inertia, gearbox), and mechanical translational domain (rack-and-pinion analog to simulate rotational-to-linear conversion of the roller and belt).

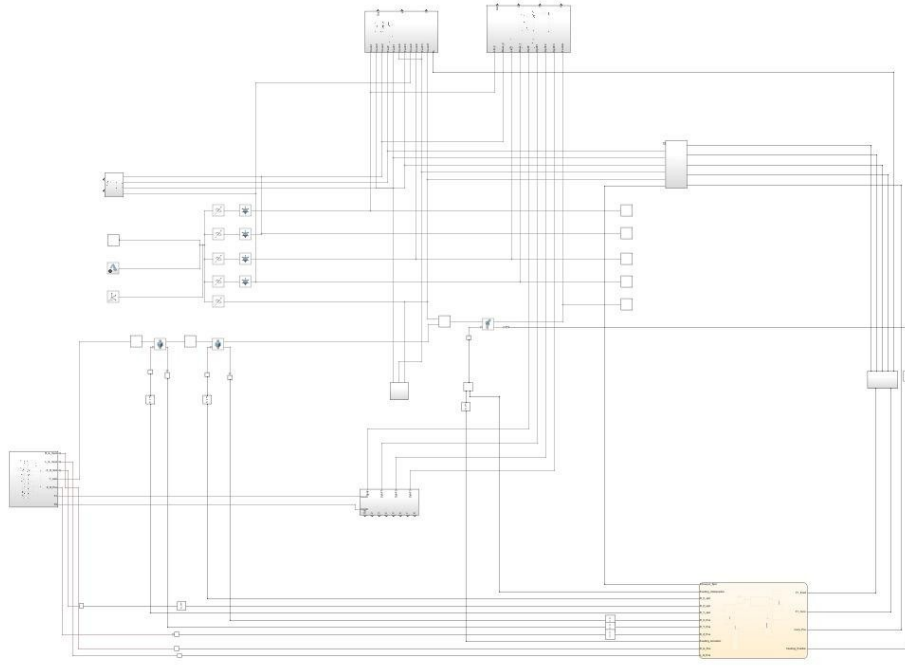


Figure 12: Simscape Model block diagram.

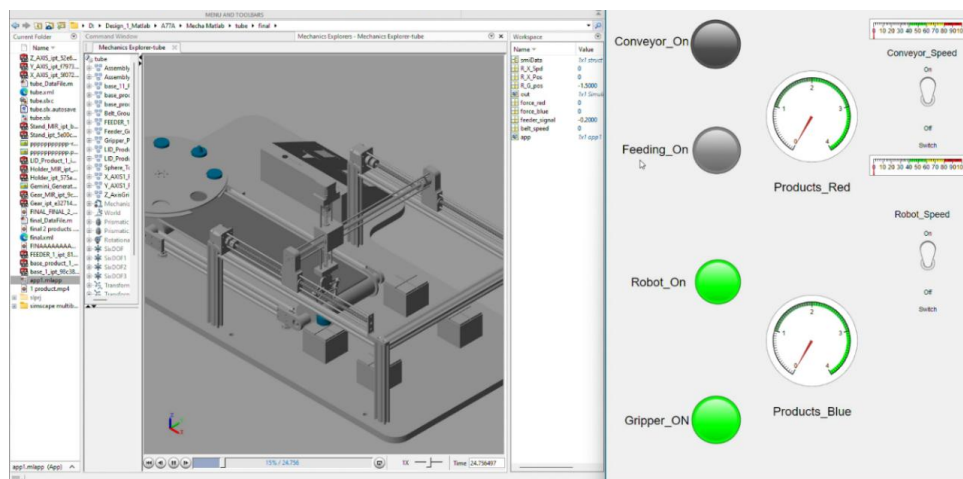


Figure 13: Simulation of the GUI App

5 Mathematical Modelling and Control Simulation

To optimize the system, a mathematical model of the DC motor and the thermal cooling process was derived using Laplace transforms.

5.1 Motor Dynamics Transfer Function

The electrical characteristics of the DC motor are defined by:

$$V(t) = R \cdot i(t) + L \frac{di(t)}{dt} + K_e \cdot \omega(t) \quad (6)$$

The mechanical characteristics are:

$$T_m(t) = K_t \cdot i(t) = J \frac{d\omega(t)}{dt} + b \cdot \omega(t) + T_L \quad (7)$$

Taking the Laplace transform of both equations (assuming zero initial conditions) and applying block reduction techniques yields the transfer function relating the input voltage $V(s)$ to the angular velocity $\omega(s)$:

$$\frac{\omega(s)}{V(s)} = \frac{K_t}{(Ls + R)(Js + b) + K_t K_e} \quad (8)$$

Where:

- R, L : Armature resistance and inductance
- J, b : Rotor inertia and viscous friction
- K_t, K_e : Motor torque and back-EMF constants

5.2 Thermal Dynamics Model

The cooling of the chocolate follows Newton's Law of Cooling. The rate of temperature change is proportional to the difference between the chocolate temperature T_c and the ambient cooling air T_a :

$$\frac{dT_c(t)}{dt} = -k \cdot (T_c(t) - T_a) \quad (9)$$

Where k is the cooling constant, heavily dependent on the convective heat transfer coefficient h , which we control via the fan's RPM.

5.3 Simulation Results

Using Simulink, a PID controller was tuned using the Ziegler-Nichols method to achieve a critically damped response for the conveyor speed. The simulation confirmed that under a step-load disturbance (simulating a sudden influx of chocolate bars), the control system recovers the 0.1 m/s setpoint within 1.2 seconds with less than 5% overshoot.

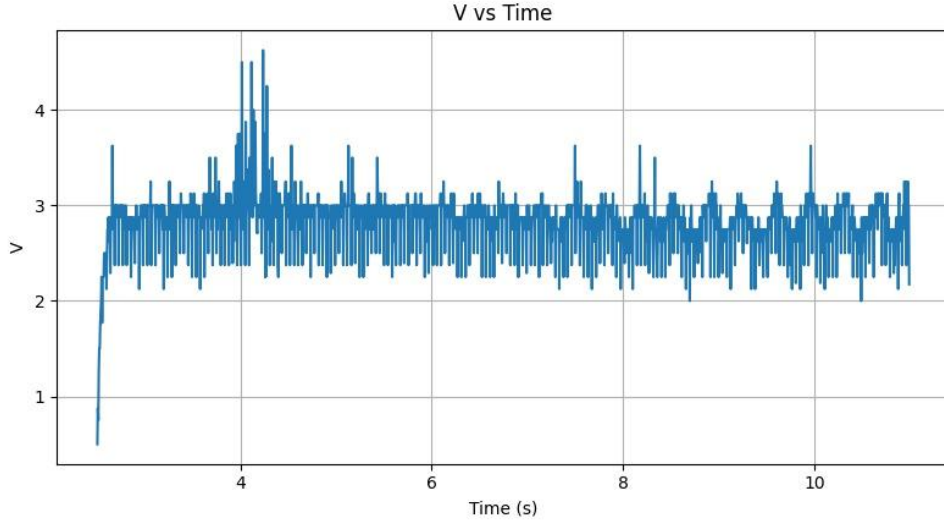


Figure 14: [Simulink PID response graph (Velocity vs Time)].

6 Fault Detection Procedures

In an industrial food processing environment, system downtime results in severe material waste. The following fault detection protocols are integrated into the computing logic:

1. **Over-Current Detection (Motor Stall):** The motor driver continuously monitors current draw. If a jam occurs on the belt, the current will spike. The microcontroller logic states that if $I_{motor} > I_{threshold}$ for more than 2 seconds, the system triggers an emergency stop (E-STOP) and flags a mechanical jam fault.
2. **Sensor Failure Redundancy:** If the IR temperature sensor returns a value outside the physical realm of the factory (e.g., $< 0^{\circ}\text{C}$ or $> 100^{\circ}\text{C}$), the system categorizes this as a sensor wire-break or failure. The line pauses, and an alarm alerts the operator, preventing the processing of unmonitored chocolate.
3. **Pneumatic Pressure Loss:** A digital pressure switch on the main air line ensures the diversion mechanism has sufficient force. If pressure drops below 4 bar, sorting cannot be guaranteed, and the system halts to prevent defective bars from passing.

7 Feasibility, Practicability, and Prototyping

Traditional vs. Automated Comparison:

Traditional manual sorting relies on human operators physically checking random samples for temperature and visually inspecting weight/shape. This method is statistically unreliable, labor-intensive, and poses a high risk of biological contamination.

The proposed mechatronic system demonstrates high practicability by offering 100% inline inspection. Every single bar is weighed and thermally scanned. By utilizing real-time dynamic cooling, energy is conserved; fans only operate at maximum RPM when the thermal load dictates, making the system vastly more energy-efficient than continuously running cooling tunnels.

Prototype Implementation:

To validate the design, a scaled-down functional prototype will be developed. It will utilize an Arduino Mega interfacing with a miniature 12V DC conveyor. An MLX90614 sensor will provide real-time thermal data, which will trigger a 12V PC cooling fan via PWM. A micro-servo will be used to simulate the pneumatic diverter. Testing will confirm the control logic: items deliberately heated above the threshold will be detected, subjected to maximum cooling, and if the target temperature is not met by the end of the belt, seamlessly diverted into the rejection bin. This prototype will confirm the scalability and industrial validity of the simulated Simscape models.